Predictive Models in ICU Databases: A Machine Learning Analysis on ICU Outcome Prediction

Mariah Noelle Cornelio, Darlene Eligado, Alain Areeba Siddiqui

DATA 4382: Data Capstone Project 2

Project Advisor: Dr. Masoud Rostami

March 6, 2026

CRITERIA BASED ON RUBRIC:

1. Clarity and Structure
2. Progress and Milestones
3. Problem-Solving and Challenges
4. Technical Depth and Accuracy
5. Future Plans and Goals

Pink = Alain, Blue = Dar, Purple = Mars

Quick Capstone 1 Recap

Objective

Data Collection

Data Understanding

During Capstone I, we recognized that developing an ICU risk prediction model requires strong domain understanding. Medical terminology and clinical context are critical, as misinterpretation of variables in this setting can have serious consequences and sometimes fatal outcomes. Because not all team members had prior clinical experience, we prioritized building foundational knowledge of ICU-related terms and variables. We consulted a practicing nurse to clarify clinical meanings and created a shared vocabulary reference sheet for all model features. This domain

grounding ensured that: clinically meaningful features were selected, variables were interpreted correctly, and model outputs remained relevant and safe for patient care contexts.

Data Quality Understanding

EDA

Data Preprocessing

Baseline with Val/Train/Test

For our baseline approach, we used a 70/15/15 train–validation–test split, with mortality defined as the positive class (1) and survival as the negative class (0). The training and validation sets were used for model development and tuning, while the test set remained untouched for unbiased evaluation. Model performance was assessed primarily through discrimination (ROC-AUC) and recall for mortality, as missing a high-risk patient carries significant clinical consequences.

Logistic Regression Baseline Model			
	Precision	**Recall**	**F1**
0	0.89	0.99	0.94
1	0.64	0.15	0.24
Macro Avg	0.77	0.57	0.59

Figure 1. Logistic Regression Baseline Model Performance Metrics by Outcome Class

The baseline results revealed a clear imbalance in performance across classes. The model performed very well in identifying survivors but struggled to correctly identify patients who died. In practical terms, it was biased toward predicting survival and failed to detect most high-risk cases. Clinically, this level of sensitivity to mortality was insufficient, highlighting the impact of class imbalance and motivating further refinement of the modeling strategy.

Modeling and Iterative Improvement

During Capstone I, we reframed our objective from bias analysis to predicting poor ICU outcomes, requiring a full revision of our modeling pipeline with a clearer clinical focus on identifying high-risk patients. To address class imbalance and better capture clinically meaningful deterioration, we consolidated readmission, hospital death, and ICU death into a single *bad_outcomes* category, increasing positive cases from 90 to 293 (relative to 2,227 negatives). We then rebuilt the dataset from the original SQL source, aggregating time-based features, adding comorbidities and intake/output variables, and computing 24-hour vital sign summaries. Most critically, we removed sources of data leakage to ensure valid prediction. This reconstruction improved both the methodological rigor and reliability of our modeling approach.

Splitting Into 3 Objectives

To better understand model behavior and data characteristics, we structured our analysis into three parallel modeling pathways. Although developed independently, each pathway was designed to address different methodological considerations and ultimately converge toward the same predictive objective for our deployment model: accurate identification of poor ICU

outcomes. A central challenge throughout this process was significant class imbalance, as adverse events were substantially less frequent than survival outcomes. This imbalance required careful evaluation to ensure the model did not become biased toward the majority class.

Capstone 1 Findings → I guess if needed, turn this into something shorter ???

Goal 1

For Goal 1, we evaluated two data scenarios: one using the original imbalanced dataset and one applying class balancing, while keeping the decision threshold fixed at 0.5. An 80/20 train–test split was used for consistent evaluation. Six models were tested under each scenario (twelve total), capturing both linear and nonlinear relationships. Under the imbalanced setting, XGBoost performed best, reflecting its ability to handle rare events effectively. Under the balanced setting, a stacked ensemble (logistic regression and XGBoost as base learners, logistic regression as meta-learner) achieved the strongest performance.

Goal 2

Goal 3

Capstone 2 Work So Far: Finding the Best Model for Deployment - pls feel free to add anything you want to talk about here

- Combining findings from Capstone 1
    - Building on findings from Capstone I, our next objective was to identify the most appropriate model for deployment. Specifically, we compared the top-performing approaches from each data scenario: the stacked model trained with class balancing and the XGBoost model trained without class balancing. Both models were further evaluated using probability calibration and threshold tuning to ensure reliable risk estimates and clinically meaningful decision boundaries.

Rationale

Our selected deployment model was chosen based on its ability to balance strong discrimination, clinical realism, and interpretability while maintaining real-world applicability.

- Strong ROC for discrimination
    - The model demonstrated strong ROC performance, indicating reliable discrimination between high- and low-risk ICU patients.
- Maintains real-world prevalence
    - Class_weight=”balanced” to improve learning of rare events, explain what this does and why it is optional
        - To preserve real-world prevalence, we avoided altering the underlying class distribution; however, class_weight="balanced"

was applied during training to help the model pay greater attention to rare adverse events. This technique increases the penalty for misclassifying minority cases, improving learning without modifying the dataset itself. Its use is optional because it adjusts training emphasis rather than the data distribution, and alternative strategies (like threshold tuning) can also address imbalance.

- Probability calibration applied
  - What is calibration - We used Isotonic
  - Ensures predicted risk % reflects real ICU outcome rates
  - Probability calibration was applied to both models using isotonic regression to ensure predicted risk probabilities accurately reflected true ICU outcome rates. Calibration aligns predicted percentages with observed event frequencies, making risk scores clinically meaningful.
- Threshold chosen for clinical goals
  - Recall: catch high risk patients
  - Acceptable precision: avoid excessive false alarms
  - A decision threshold was selected based on clinical priorities: prioritizing recall to capture high-risk patients while maintaining acceptable precision to avoid excessive false alarms.
- Explainability integrated for transparency per patient

- Finally, explainability methods were integrated to provide transparent, patient-level insights, supporting clinician trust and responsible deployment.

Model Training Strategies

Two candidate models were developed and evaluated: an XGBoost model trained without class balancing to preserve real-world outcome prevalence, and a stacked model trained with class_weight="balanced" to improve learning of rare adverse events during optimization. Both models utilized the reconstructed Capstone I dataset consisting of engineered clinical features, including comorbidities, intake/output variables, and 24-hour summary statistics for vital signs. The shape of this final data set was 2,520 rows with 55 features.

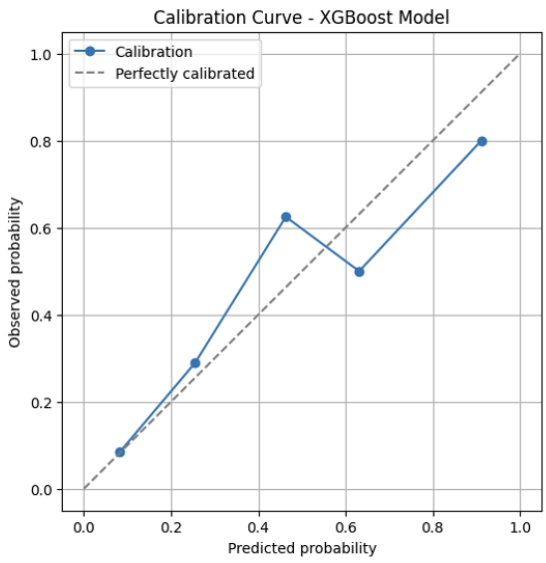
The data was partitioned using an 80/20 train-test split. Within the training set, 5-fold cross-validation with out-of-fold (OOF) predictions and grid search was applied to optimize ROC-AUC and identify the best-performing hyperparameter configuration. The selected models were then calibrated using isotonic regression to improve probability reliability before being evaluated on the untouched hold-out test set to assess generalization performance. Following validation, each model was prepared for deployment through clinically informed threshold selection and integration of SHAP-based explainability to provide transparent, patient-level risk interpretation.

Model Metric Comparison

Table 1. Performance Metrics Supporting Final Deployment Model Selection

Comparative Model Evaluation							
Model	**Class Weight Handling**	**Calibrated Test ROC**	**Calibrated Test PRC**	**Calibrated Brier Score**	**Recall**	**Precision**	**Specificity**
Stacked	Balanced	0.814	0.40	0.0943	0.627	0.356	0.849
XGBoost	No balance	0.775	0.379	0.0897	0.797	0.229	0.645

The stacked model trained with class balancing achieved a higher calibrated ROC-AUC (0.814 vs. 0.775) and PR-AUC (0.40 vs. 0.379) compared to XGBoost, indicating stronger overall discrimination and slightly better performance in identifying positive cases across thresholds. It also demonstrated higher specificity (0.849 vs. 0.645) and improved precision (0.356 vs. 0.229), meaning it generated fewer false positives and more reliable positive predictions. In contrast, the unbalanced XGBoost model achieved substantially higher recall (0.797 vs. 0.627), indicating greater sensitivity in capturing high-risk ICU patients. However, this came at the cost of lower precision and specificity, reflecting an increased false alarm rate.



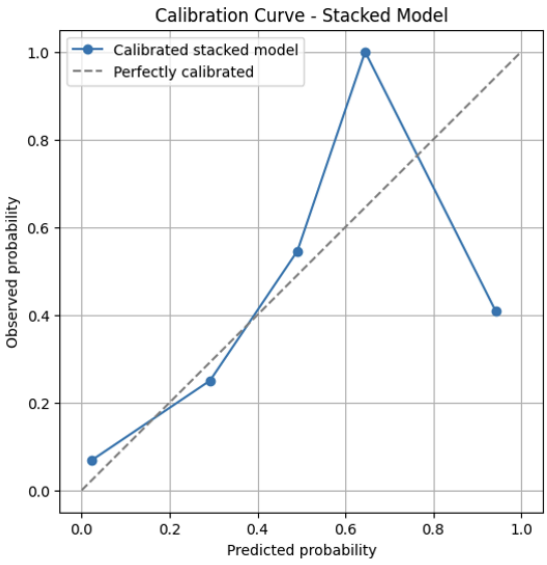


Figure 2. Probability Calibration Performance of Candidate Deployment Models, XGBoost (left) and Stacked Ensemble (right)

The Brier score measures the accuracy of predicted probabilities by calculating the mean squared difference between predicted risk and actual outcomes. Lower values indicate better-calibrated and more accurate probability estimates. While both models performed similarly, XGBoost achieved a slightly lower (better) Brier score (0.0897 vs. 0.0943), suggesting marginally more accurate probability calibration overall. Together, these results highlight a trade-off between sensitivity and false-positive control, requiring final model selection to align with clinical deployment priorities.

Global and Local SHAP on Stacked Model

- Test Patient SHAP

XAI-First ICU Risk Interface Prototype

Future Plans and Goals - pls feel free to add anything you want to talk about here

- Try and create the best model with these considerations
    - Recall goal 0.70, precision goal 0.70
    - Good PRC and ROC curve (ideally 0.80)
    - For validation do stratified K folding or nested
    - Don’t call it imbalanced/balanced dataset but rather handling balance strategy

    - For stacked model try SVM and XGB to capture nonlinear relationships
        - Look into what Landon was saying about transparency and SHAP on whole model
    - Try SMOTE/synthetic data on training data
    - Calibrate
    - Adaptive learning

Future development will focus on optimizing a model that achieves stronger clinical balance, targeting both recall and precision of approximately 0.70 while maintaining robust discrimination (ROC-AUC and PR-AUC ideally greater than or equal to 0.80). Validation strategies will be strengthened using stratified k-fold or nested cross-validation to ensure more reliable generalization estimates. Rather than framing the dataset itself as imbalanced, future iterations will refine the balance-handling strategy during training. For the stacked architecture, incorporating models such as SVM and XGBoost may better capture nonlinear relationships. Synthetic sampling techniques (like SMOTE) will also be explored within the training set only to enhance minority class learning. Further work will evaluate model-level transparency, including SHAP explanations across the full ensemble as suggested, alongside continued probability calibration to preserve reliable risk estimates. Finally, adaptive learning strategies may be considered to allow model performance to evolve with changing ICU data patterns over time.

- Deployment model - ICU risk alert system
    - What is the goal for this; balance recall and precision

- Patient first 24 hour vitals and demographics and history of comorbidities as inputs
- Output as this: Patient ID, bad outcome risk %, Risk contributors (Local SHAP values), recommended course of action (based on relative range and triage actions - Immediate ICU attention, standard monitoring, monitor closely), and Actual Outcome to see how model performed, but this won't be on deployment model

	Patient	Bad Outcome Risk (%)	Reasons Why	Recommended Course of Action	Actual Outcome
0	1075	95.300000	heartrate_std (increases risk); sao2_mean (increases risk); respiration_mean (increases risk)	Immediate ICU attention	1
1	1401	0.000000	age (decreases risk); apachescore (decreases risk); nettotal (decreases risk)	Standard monitoring	0
2	423	0.700000	age (decreases risk); activetx (decreases risk); meanbp (increases risk)	Standard monitoring	0
3	603	0.200000	nettotal (decreases risk); activetx (decreases risk); bun (increases risk)	Standard monitoring	1
4	837	6.200000	activetx (increases risk); nettotal (increases risk); meanbp (decreases risk)	Monitor closely / early intervention	1
5	595	0.000000	nettotal (decreases risk); activetx (decreases risk); sao2_mean (decreases risk)	Standard monitoring	0
6	1899	0.000000	apachescore (decreases risk); activetx (decreases risk); heartrate_mean (decreases risk)	Standard monitoring	0
7	424	0.000000	apachescore (decreases risk); age (increases risk); nettotal (decreases risk)	Standard monitoring	0
8	746	100.000000	ph (increases risk); meanbp (increases risk); sao2_mean (increases risk)	Immediate ICU attention	1
9	166	1.400000	apachescore (decreases risk); age (increases risk); sao2_mean (increases risk)	Standard monitoring	0

Figure 3. Example prototype deployment of ICU risk alert system

The proposed deployment model functions as an ICU risk alert system designed to support early clinical decision-making. Its primary goal is to balance recall and precision—ensuring high-risk patients are identified promptly while minimizing excessive false alarms that could contribute to alert fatigue. Model inputs consist of patient data from the first 24 hours of ICU admission, including vital sign summary statistics, demographic information, and comorbidity history. The system outputs a structured risk report containing: Patient ID, predicted bad outcome risk percentage, key risk contributors (via local SHAP values), and a recommended course of action based on predefined risk thresholds (Immediate ICU attention, Monitor closely, Standard monitoring). For evaluation purposes, actual outcomes will be compared against predictions to

assess model performance; however, true outcomes will not be visible within the live deployment interface.

Conclusion - Im guessing just summarize everything and what we hope to achieve?